

$$\nabla^2 \mathbf{E} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = 0,$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \cong 3 \times 10^8 \text{ (m/s)} = 300 \text{ (Mm/s)}$$

$$\nabla^2 \mathbf{E} + k_0^2 \mathbf{E} = 0,$$

$$k_0 = \omega \sqrt{\mu_0 \epsilon_0} = \frac{\omega}{c}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k_0^2 \right) E_x = 0.$$

$$\frac{d^2 E_x}{dz^2} + k_0^2 E_x = 0,$$

$$\begin{aligned} E_x(z) &= E_x^+(z) + E_x^-(z) \\ &= E_0^+ e^{-jk_0 z} + E_0^- e^{jk_0 z} \end{aligned}$$

$$\begin{aligned} E_x^+(z, t) &= \Re_e[E_x^+(z)e^{j\omega t}] \\ &= \Re_e[E_0^+ e^{j(\omega t - k_0 z)}] \\ &= E_0^+ \cos(\omega t - k_0 z) \end{aligned}$$

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}$$

$$\nabla \times \mathbf{E} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 0 & 0 & \frac{\partial}{\partial z} \\ E_x^+(z) & 0 & 0 \end{vmatrix} = -j\omega\mu_0(\mathbf{a}_x H_x^+ + \mathbf{a}_y H_y^+ + \mathbf{a}_z H_z^+)$$

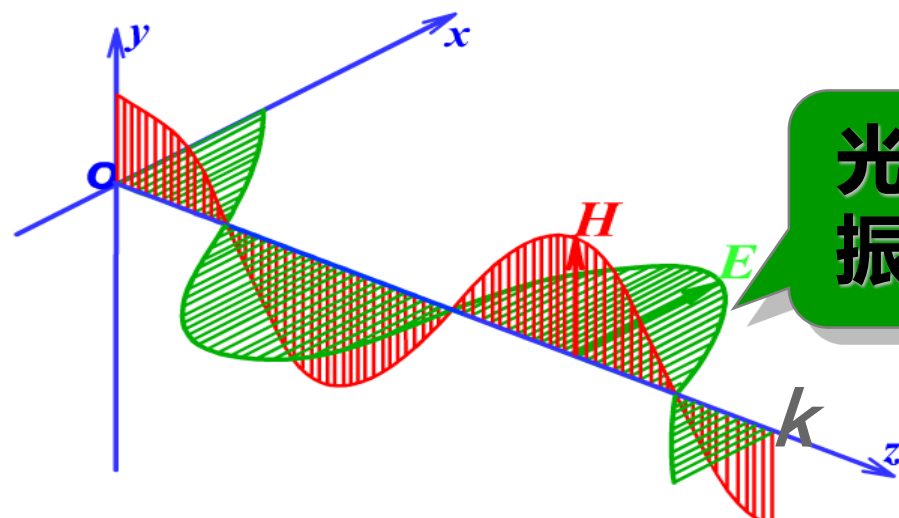
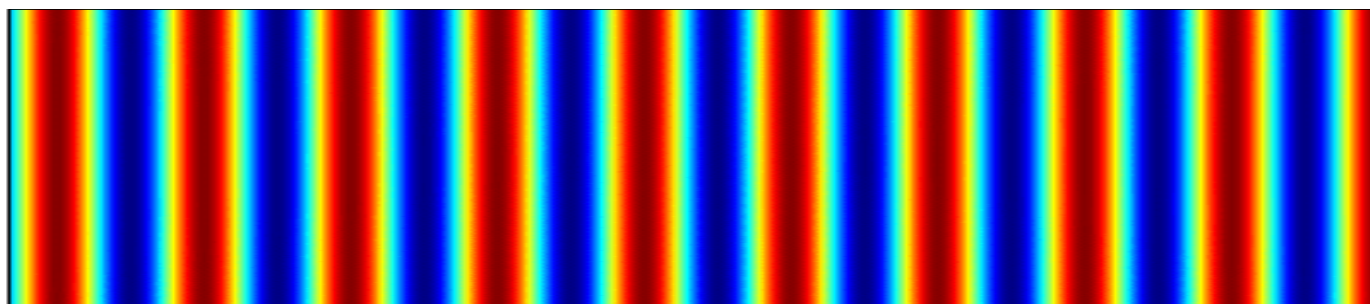
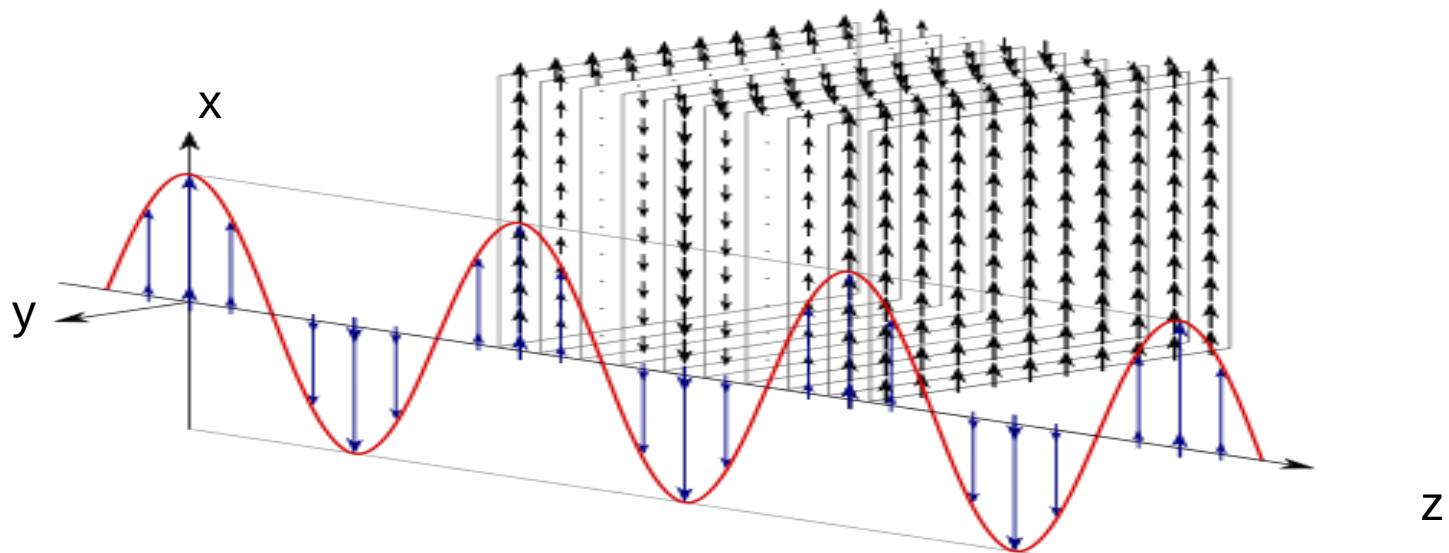
$$H_x^+ = 0,$$

$$H_y^+ = \frac{1}{-j\omega\mu_0} \frac{\partial E_x^+(z)}{\partial z},$$

$$H_z^+ = 0.$$

$$= \frac{k_0}{\omega\mu_0} E_x^+(z) = \frac{1}{\eta_0} E_x^+(z)$$

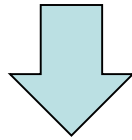
$$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} \cong 120\pi \cong 377 \quad (\Omega)$$



光矢量
振动面

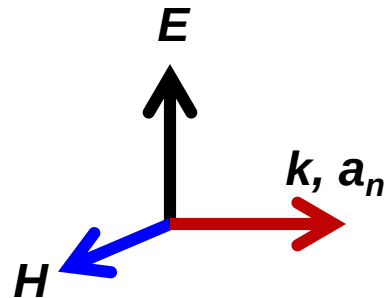
$$\mathbf{E}(x, y, z) = \mathbf{E}_0 e^{-jk_x x - jk_y y - jk_z z} = \mathbf{E}_0 e^{-j\mathbf{k} \cdot \mathbf{R}} = \mathbf{E}_0 e^{-jk\mathbf{a}_n \cdot \mathbf{R}}$$

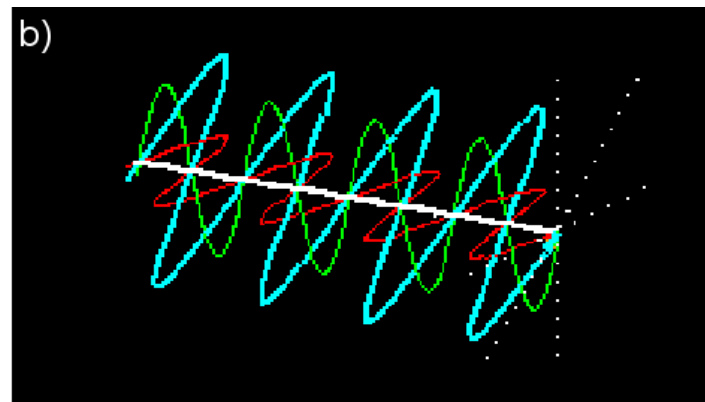
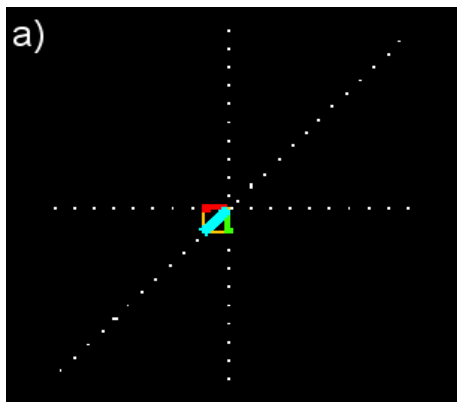
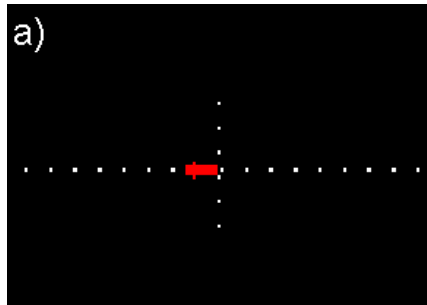
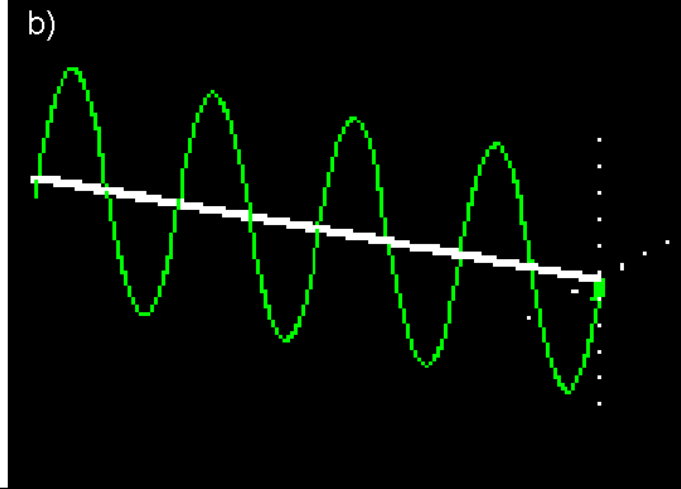
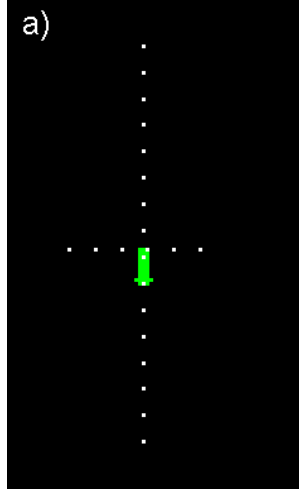
$$\mathbf{H}(\mathbf{R}) = -\frac{1}{j\omega\mu} \nabla \times \mathbf{E}(\mathbf{R})$$



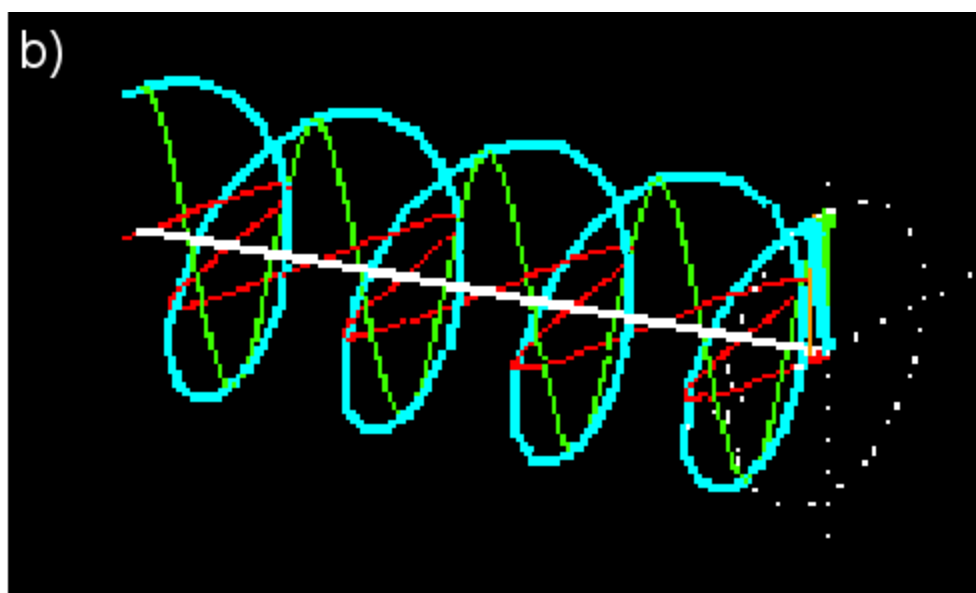
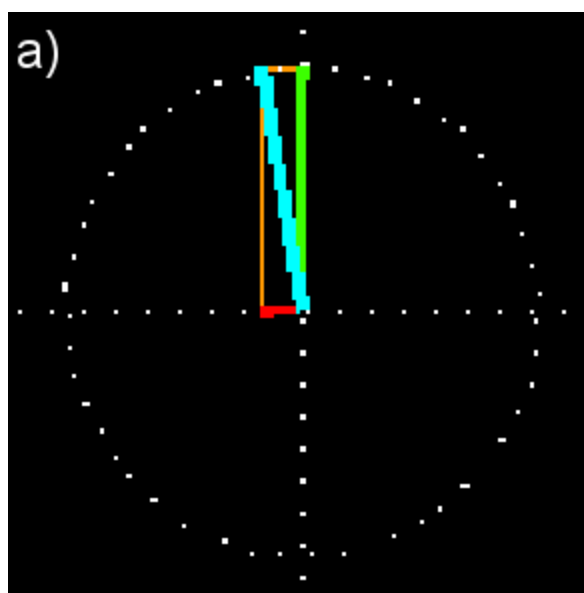
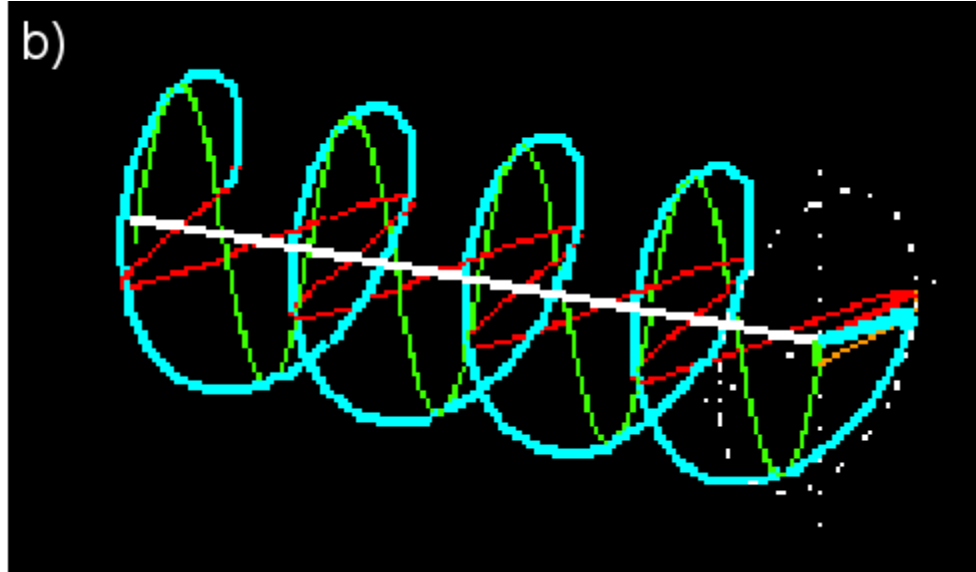
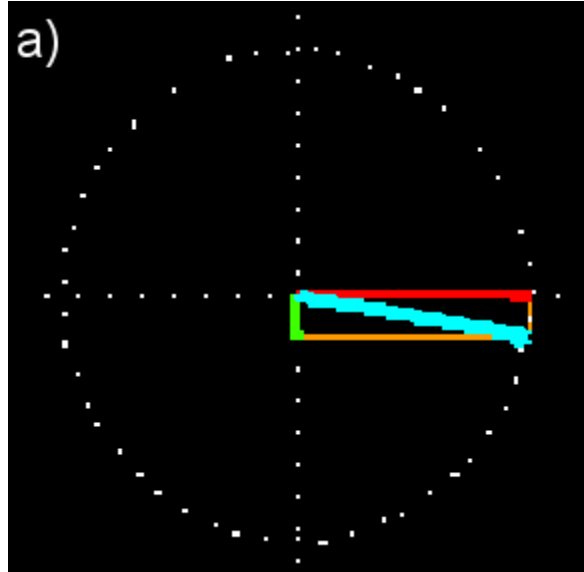
$$\mathbf{E}(\mathbf{R}) = \mathbf{E}_0 e^{-j\mathbf{k} \cdot \mathbf{R}} = \underline{\mathbf{E}_0 e^{-jk\mathbf{a}_n \cdot \mathbf{R}}}$$

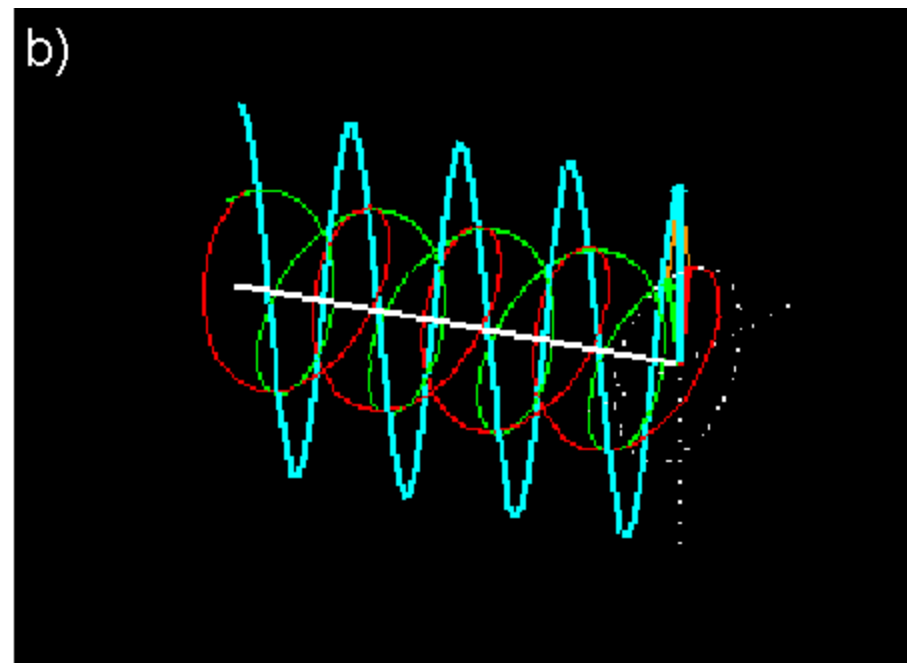
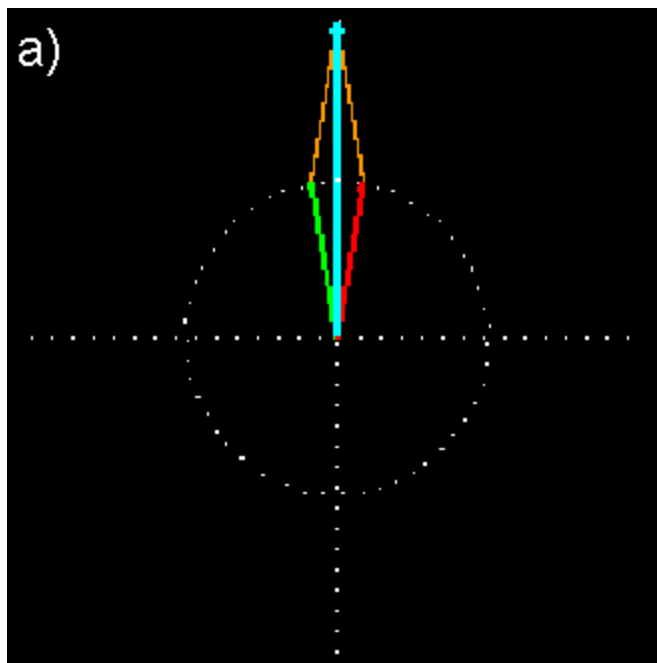
$$\mathbf{H}(\mathbf{R}) = \frac{1}{\eta} \mathbf{a}_n \times \mathbf{E}(\mathbf{R})$$





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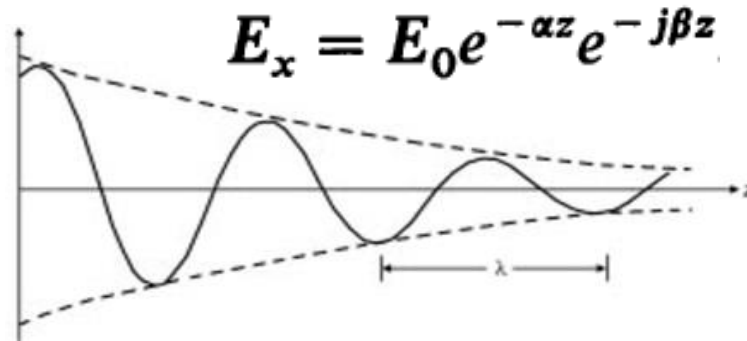




§ 8-3 Plane Waves in Lossy Media

$$\nabla^2 \mathbf{E} + k_c^2 \mathbf{E} = 0$$

$k_c = \omega \sqrt{\mu \epsilon_c}$ is a complex number.



$$\begin{aligned} \epsilon_c &= \epsilon' - j\epsilon'' \\ &= \epsilon - j\frac{\sigma}{\omega} \end{aligned}$$

Low-Loss Dielectrics

$$\beta \cong \omega \sqrt{\mu \epsilon'} \left[1 + \frac{1}{8} \left(\frac{\epsilon''}{\epsilon'} \right)^2 \right]$$

$$\gamma = \alpha + j\beta \cong j\omega \sqrt{\mu \epsilon'} \left[1 - j \frac{\epsilon''}{2\epsilon'} + \frac{1}{8} \left(\frac{\epsilon''}{\epsilon'} \right)^2 \right], \quad \alpha \cong \frac{\omega \epsilon''}{2} \sqrt{\frac{\mu}{\epsilon'}}$$

Good Conductors

$$\gamma = \alpha + j\beta = j\omega \sqrt{\mu \epsilon} \left(1 + \frac{\sigma}{j\omega \epsilon} \right)^{1/2}$$

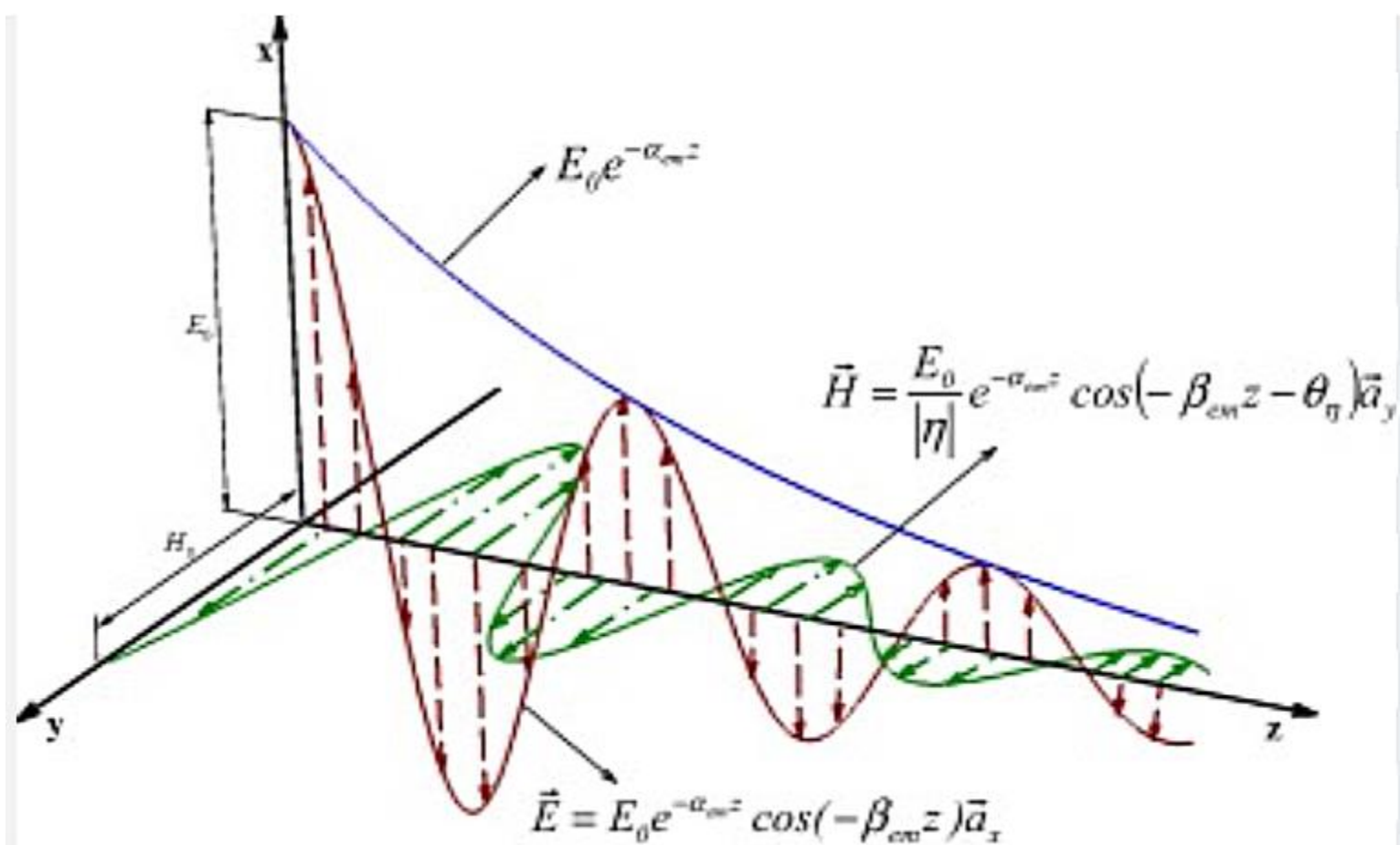
$$\gamma = \alpha + j\beta \cong (1 + j) \sqrt{\pi f \mu \sigma}$$

$$\lambda = \frac{2\pi}{\beta} = 2 \sqrt{\frac{\pi}{f \mu \sigma}}$$

$$\eta_c = \sqrt{\frac{\mu}{\epsilon_c}} \cong \sqrt{\frac{j\omega \mu}{\sigma}} = (1 + j) \sqrt{\frac{\pi f \mu}{\sigma}}$$

➔ **H lags behind E by 45°.**

$$\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}}$$



(3) Phase velocity

$$u_p = \frac{\omega}{\beta} \cong \sqrt{\frac{2\omega}{\mu\sigma}}$$

(4) wavelength

$$\lambda = \frac{2\pi}{\beta} = 2 \sqrt{\frac{\pi}{f\mu\sigma}}$$

Copper @ 3 MHz $\lambda_0=100\text{m}$

$$= 720 \text{ (m/s)}$$

$$= 0.24 \text{ (mm)}$$

(5) **Skin depth (趋肤深度) or depth of penetration (穿透深度)**

$$\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

At microwave frequencies, the skin depth is **0.66 μm (at 10 GHz for copper)**. Thus, the fields and the electric currents can be considered as confined in a very thin layer of the conductor surface (Skin Effect).

Skin depth of good metals

$$\delta = \frac{1}{\alpha} = \frac{1}{\sqrt{\pi f \mu \sigma}}$$

Skin Depths, δ in (mm), of Various Materials

Material	σ (S/m)	$f = 60$ (Hz)	1 (MHz)	1 (GHz)
Silver	6.17×10^7	8.27 (mm)	0.064 (mm)	0.0020 (mm)
Copper	5.80×10^7	8.53	0.066	0.0021
Gold	4.10×10^7	10.14	0.079	0.0025
Aluminum	3.54×10^7	10.92	0.084	0.0027
Iron ($\mu_r \cong 10^3$)	1.00×10^7	0.65	0.005	0.00016
Seawater	4	32 (m)	0.25 (m)	†

† The ϵ of seawater is approximately $72\epsilon_0$. At $f = 1$ (GHz), $\sigma/\omega\epsilon \cong 1$ (not $\gg 1$). Under these conditions, seawater is not a good conductor, and Eq. (8-57) is no longer applicable.

8-4 The electric field intensity of a linearly polarized uniform plane wave propagating in the +z-direction in seawater is $\vec{E} = \vec{a}_x 100 \cos(10^7 \pi t)$ (V / m) at $\underline{z=0}$. The constitutive parameters of seawater are $\epsilon_r=72$, $\mu_r=1$, $\sigma=4$ (S/m)

(a) Determine the attenuation constant, phase constant, intrinsic impedance, phase velocity, wavelength, and skin depth.

(b) Find the distance at which the amplitude of E is 1% of its value at $\underline{z=0}$.

(c) Write the expressions for $\vec{E}(z,t)$ and $\vec{H}(z,t)$ at $\underline{z=0.8}$ (m) as functions of t .

$$\omega = 10^7 \pi$$

$$f = \frac{\omega}{2\pi} = 5 \times 10^6$$

$$\frac{\sigma}{\omega \epsilon} = \frac{\sigma}{\omega \epsilon_0 \epsilon_r} = \frac{4}{10^7 \pi \left(\frac{1}{36\pi} \times 10^{-9} \right) 72} = 200 \gg 1.$$

a) Attenuation constant:

$$\alpha = \sqrt{\pi f \mu \sigma} = \sqrt{5\pi 10^6 (4\pi 10^{-7}) 4} = 8.89.$$

Phase constant:

$$\beta = \sqrt{\pi f \mu \sigma} = 8.89$$

Intrinsic impedance:

$$\eta_c = \sqrt{\frac{\mu}{\epsilon_c}} \approx \sqrt{-\mu \frac{\omega}{j\sigma}} = \sqrt{j\mu \frac{\omega}{\sigma}} = \pi e^{j\pi/4} \quad (\Omega)$$

Phase velocity:

$$u_p = \frac{\omega}{\beta} = \frac{10^7 \pi}{8.89} = 3.53 \times 10^6 \quad (m/s).$$

Wavelength:

$$\lambda = \frac{2\pi}{\beta} = \frac{2\pi}{8.89} = 0.707 \quad (m)$$

Skin depth:

$$\delta = \frac{1}{\alpha} = \frac{1}{8.89} = 0.112 \quad (m).$$

b) Distance z_1 at which the amplitude of wave decreases to 1% of its value at $z=0$:

$$e^{-\alpha z_1} = 0.01 \quad \text{or} \quad e^{\alpha z_1} = \frac{1}{0.01} = 100$$

$$z_1 = \frac{1}{\alpha} \ln 100 = \frac{4.605}{8.89} = 0.518 \quad (m)$$

c) The instantaneous expression for $\vec{E}$ is

$$\vec{E}(z,t) = \Re[\vec{E}(z)e^{j\omega t}] = \Re[\vec{a}_x 100e^{-\alpha z} e^{j(\omega t - \beta z)}] = \vec{a}_x 100e^{-\alpha z} \cos(\omega t - \beta z).$$

At $z=0.8(\text{m})$ we have

$$\vec{E}(0.8,t) = \vec{a}_x 100e^{-0.8\alpha} \cos(10^7 \pi t - 0.8\beta) = \vec{a}_x 0.082 \cos(10^7 \pi t - 7.11) \quad (\text{V} / \text{m})$$

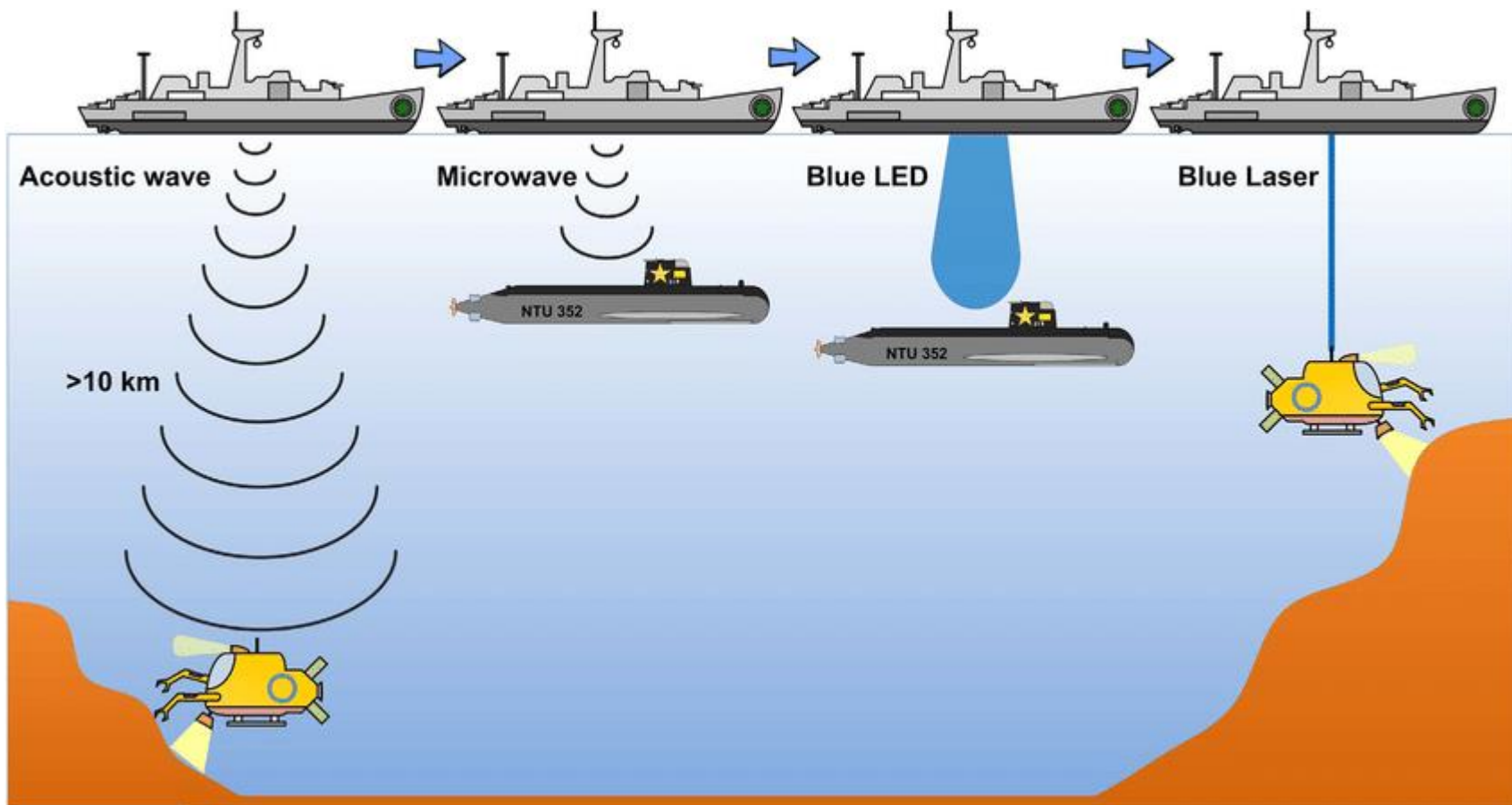
$$H_y(z) = E_x(z) / \eta_c$$

$$H_y(0.8) = \frac{100e^{-0.8\alpha} e^{-j0.8\beta}}{\pi e^{j\pi/4}} = \frac{0.082e^{-j7.11}}{\pi e^{j\pi/4}} = 0.026e^{-j1.61}.$$

$$\vec{H}(0.8,t) = \vec{a}_y 0.026 \cos(10^7 \pi t - 1.61) \quad (\text{A} / \text{m}).$$

A 5(MHz) plane wave attenuates very rapidly in seawater and becomes negligibly weak a very short distance from the source (0.5m).

Even at very low frequencies, long-distance radio communication with a submerged submarine is very difficult.



8-6 A narrow-band signal propagates in a lossy dielectric medium which has a loss tangent 0.2 at 550(kHz), the carrier frequency of the signal. The dielectric constant of the medium is 2.5.

(a) Determine α and β . (b) Determine μ_p and μ_g . Is the medium dispersive?

a) Since the loss tangent $\varepsilon''/\varepsilon' = 0.2$ and $\varepsilon''^2/8\varepsilon'^2 \ll 1$

Low-loss dielectric

$$\varepsilon'' = 0.2\varepsilon' = 0.2 \times 2.5\varepsilon_0 = 4.42 \times 10^{-12}$$

$$\alpha = \frac{\omega\varepsilon''}{2} \sqrt{\frac{\mu}{\varepsilon'}} = \pi(550 \times 10^3) \times (4.42 \times 10^{-12}) \times \frac{377}{\sqrt{2.5}} = 1.82 \times 10^{-3}$$

$$\beta = \omega\sqrt{\mu\varepsilon'} \left[1 + \frac{1}{8} \left(\frac{\varepsilon''}{\varepsilon'}\right)^2\right] = 2\pi(550 \times 10^3) \frac{\sqrt{2.5}}{3 \times 10^8} \left[1 + \frac{1}{8} (0.2)^2\right] = 0.0183$$

b) Phase velocity :

$$u_p = \frac{\omega}{\beta} = \frac{1}{\sqrt{\mu\varepsilon'} \left[1 + \frac{1}{8} \left(\frac{\varepsilon''}{\varepsilon'}\right)^2\right]} \cong \frac{1}{\sqrt{\mu\varepsilon'}} \left[1 - \frac{1}{8} \left(\frac{\varepsilon''}{\varepsilon'}\right)^2\right] = 1.888 \times 10^8$$

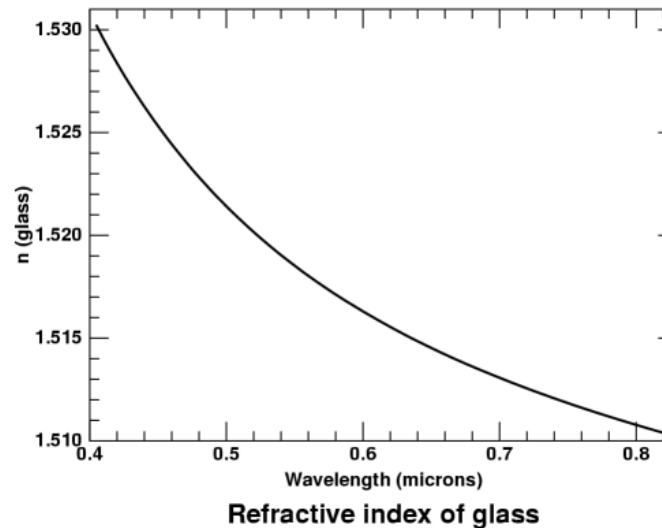
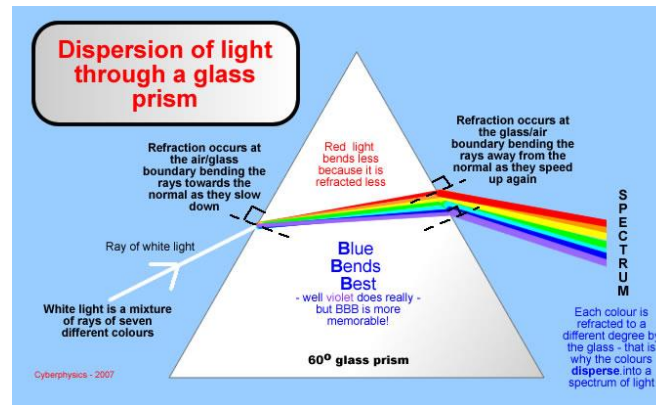
c) Group velocity

$$\frac{d\beta}{d\omega} = \sqrt{\mu\epsilon'} \left[1 + \frac{1}{8} \left(\frac{\epsilon''}{\epsilon'} \right)^2 \right].$$
$$u_g = \frac{1}{(d\beta/d\omega)} \cong \frac{1}{\sqrt{\mu\epsilon'}} \cong \mu_p.$$

Thus a low-loss dielectric is nearly nondispersive. Here we have assumed ϵ'' to be independent of frequency. For a high-loss dielectric, ϵ'' will be a function of ω and may have a magnitude comparable to ϵ' . The approximation in Eq(8-49) will no longer hold, and the medium will be dispersive.

§ 8-4 Group Velocity (群速度)

Dispersion (色散) : relationship between phase velocity (or refractive index) on frequency
 $\beta \sim \omega$ or $n \sim \omega$



§ 8-4 Group Velocity (群速度)

Phase velocity: the velocity of propagation of an equiphase wavefront.

$$u_p = \frac{dz}{dt} = \frac{\omega_0}{\beta_0} \quad \longleftarrow \omega_0 t - \beta_0 z = \text{Constant:}$$

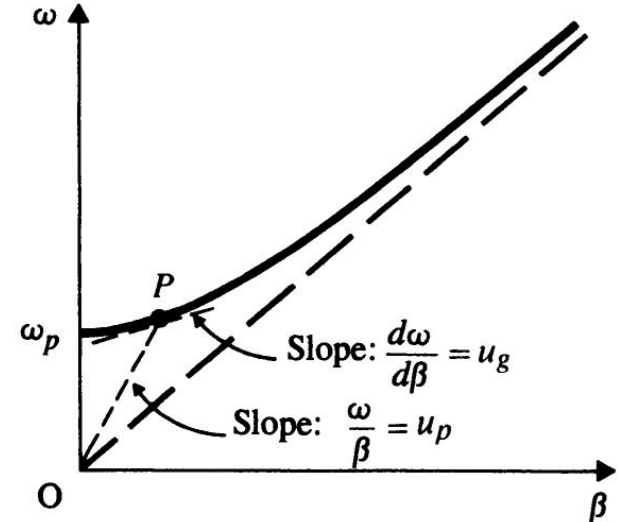
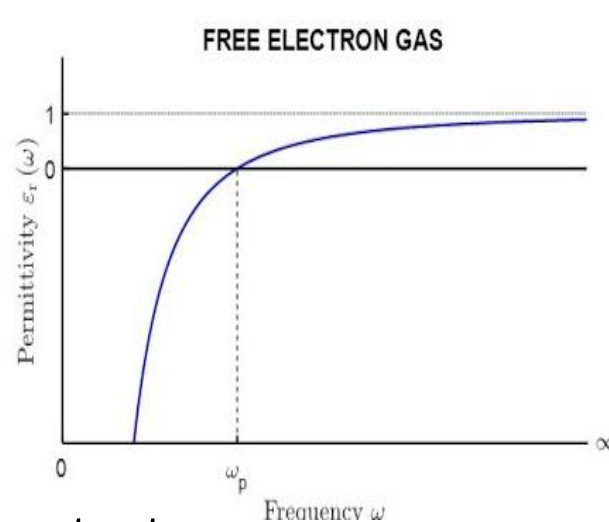
($\beta = \omega\sqrt{\mu\epsilon}$ in lossless media)

Group velocity: the velocity of propagation of a wave-packet envelope (of a group of frequencies)

$$u_g = \frac{1}{d\beta/d\omega} \quad (\text{m/s}).$$

Wave propagation in **ionized medium**

$$\epsilon_p = \epsilon_0 \left(1 - \frac{\omega_p^2}{\omega^2} \right)$$



(1) Propagation constant

$$\beta = \frac{\omega}{c} \sqrt{1 - \left(\frac{\omega_p}{\omega} \right)^2}$$

$$v \sqrt{\mu \epsilon_0} \sqrt{1 - \left(\frac{f_p}{f} \right)^2},$$



When $f < f_p$, the electromagnetic wave will decay in this plasma



When $f > f_p$, the electromagnetic wave will propagate in this plasma.

The plasma frequency f_p is also referred to as the “cutoff frequency” .

Wave propagation in **ionized medium**

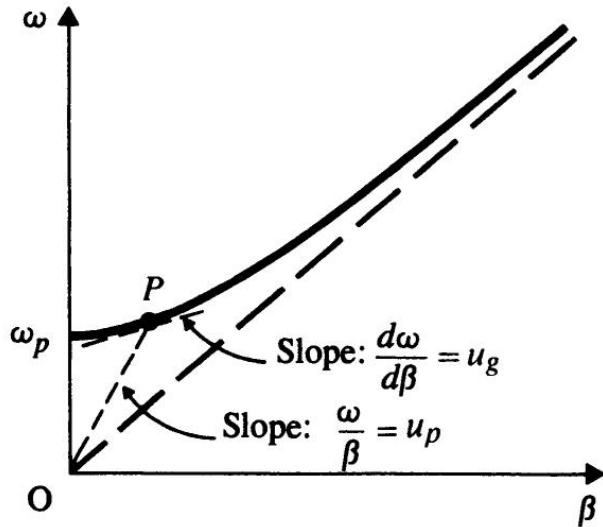


FIGURE 8-7
 ω - β graph for ionized gas.

(2) Phase velocity

$$u_p = \frac{\omega}{\beta} = \frac{c}{\sqrt{1 - \left(\frac{\omega_p}{\omega}\right)^2}}.$$

(3) Group velocity

$$u_g = c \sqrt{1 - \left(\frac{\omega_p}{\omega}\right)^2}.$$

$$u_p \geq c \text{ and } u_g \leq c.$$

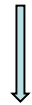
$$u_p u_g = c^2$$

§ 8-5 Flow of Electromagnetic Power and the Poynting Vector

Poynting's law (坡印廷定律) is a law of energy conservation.
It has already been involved in Maxwell's equations.

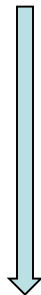
Derive the expression of Poynting's law:

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = \mathbf{H} \cdot (\nabla \times \mathbf{E}) - \mathbf{E} \cdot (\nabla \times \mathbf{H}).$$



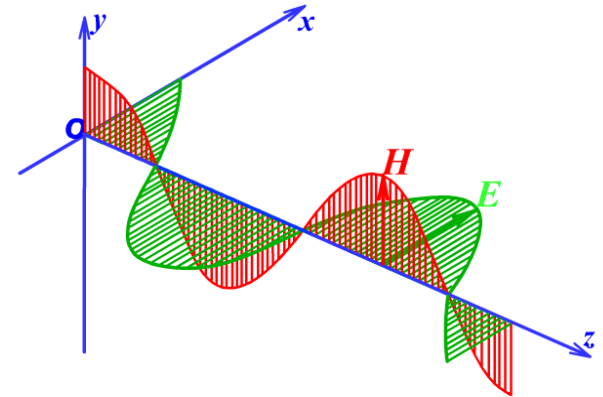
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}.$$

$$= -\mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} - \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} - \mathbf{E} \cdot \mathbf{J}.$$

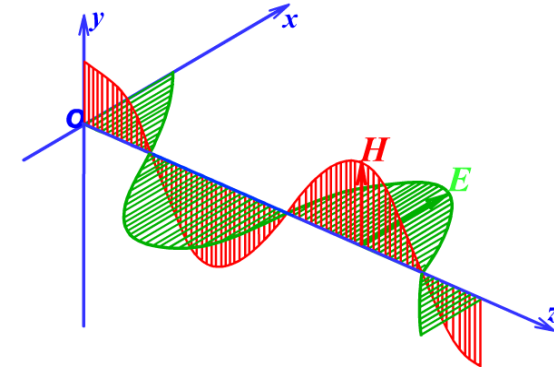


$$\begin{aligned} \mathbf{H} \cdot \frac{\partial \mathbf{B}}{\partial t} &= \mathbf{H} \cdot \frac{\partial(\mu \mathbf{H})}{\partial t} = \frac{1}{2} \frac{\partial(\mu \mathbf{H} \cdot \mathbf{H})}{\partial t} = \frac{\partial}{\partial t} \left(\frac{1}{2} \mu H^2 \right), \\ \mathbf{E} \cdot \frac{\partial \mathbf{D}}{\partial t} &= \mathbf{E} \cdot \frac{\partial(\epsilon \mathbf{E})}{\partial t} = \frac{1}{2} \frac{\partial(\epsilon \mathbf{E} \cdot \mathbf{E})}{\partial t} = \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon E^2 \right), \\ \mathbf{E} \cdot \mathbf{J} &= \mathbf{E} \cdot (\sigma \mathbf{E}) = \sigma E^2. \end{aligned}$$

$$= -\frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) - \sigma E^2,$$



Equation of Poynting's Vector



■ Differential微分 form of Poynting's law

$$\nabla \cdot (\mathbf{E} \times \mathbf{H}) = -\frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) - \sigma E^2 \quad (8-81)$$

■ Integration积分 form of Poynting's law (Poynting's theorem坡印廷定理)

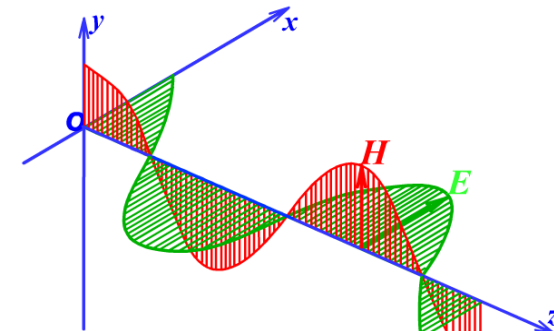
$$\oint_S (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{s} = -\frac{\partial}{\partial t} \int_V \left(\frac{1}{2} \epsilon E^2 + \frac{1}{2} \mu H^2 \right) dv - \int_V \sigma E^2 dv, \quad (8-82)$$

the surface integral of $\mathbf{P}$ over a closed surface = the power leaving the enclosed volume

Poynting vector: power flow per unit area (power density vector)

$$\mathcal{P} = \mathbf{E} \times \mathbf{H} \quad (\text{W/m}^2).$$

Poynting's theorem 坡印廷定理



$$-\oint_S \mathcal{P} \cdot d\mathbf{s} = \frac{\partial}{\partial t} \int_V (w_e + w_m) dv + \int_V p_\sigma dv, \quad (8-84)$$

where

$$w_e = \frac{1}{2}\epsilon E^2 = \frac{1}{2}\epsilon \mathbf{E} \cdot \mathbf{E}^* = \text{Electric energy density}, \quad (8-85)$$

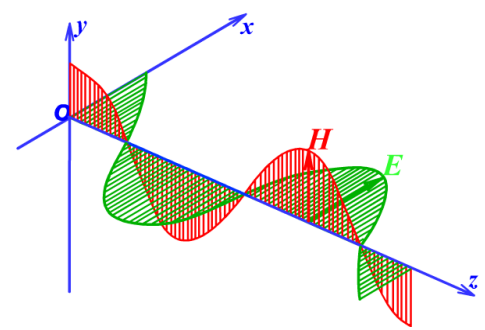
$$w_m = \frac{1}{2}\mu H^2 = \frac{1}{2}\mu \mathbf{H} \cdot \mathbf{H}^* = \text{Magnetic energy density}, \quad (8-86)$$

$$p_\sigma = \sigma E^2 = J^2/\sigma = \sigma \mathbf{E} \cdot \mathbf{E}^* = \mathbf{J} \cdot \mathbf{J}^*/\sigma = \text{Ohmic power density}. \quad (8-87)$$

In words, Eq. (8-84) states that the total power flowing *into* a closed surface at any instant equals the sum of the rates of increase of the stored electric and magnetic energies and the ohmic power dissipated within the enclosed volume.

8-5.1 INSTANTANEOUS AND AVERAGE POWER DENSITIES

$$\mathcal{P} = \mathbf{E} \times \mathbf{H} \quad (\text{W/m}^2).$$



$$\underline{\mathbf{E}(z)} = \mathbf{a}_x E_x(z) = \mathbf{a}_x E_0 e^{-(\alpha + j\beta)z}, \quad (8-88)$$

the instantaneous expression is

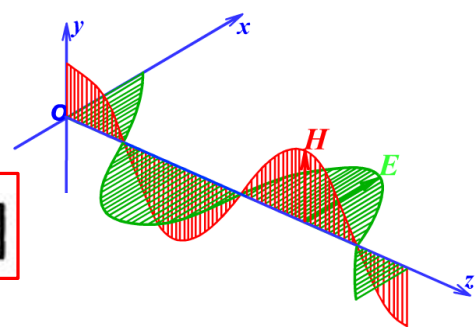
$$\begin{aligned} \underline{\mathbf{E}(z, t)} &= \Re[e[\underline{\mathbf{E}(z)}e^{j\omega t}]] = \mathbf{a}_x E_0 e^{-\alpha z} \Re[e^{j(\omega t - \beta z)}] \\ &= \mathbf{a}_x E_0 e^{-\alpha z} \cos(\omega t - \beta z). \end{aligned} \quad (8-89)$$

$$\underline{\mathbf{H}(z)} = \mathbf{a}_y H_y(z) = \mathbf{a}_y \frac{E_0}{|\eta|} e^{-\alpha z} e^{-j(\beta z + \theta_\eta)}, \quad (8-90)$$

where θ_η is the phase angle of the intrinsic impedance $\eta = |\eta|e^{j\theta_\eta}$ of the medium. The corresponding instantaneous expression for $\mathbf{H}(z)$ is

$$\underline{\mathbf{H}(z, t)} = \Re[e[\underline{\mathbf{H}(z)}e^{j\omega t}]] = \mathbf{a}_y \frac{E_0}{|\eta|} e^{-\alpha z} \cos(\omega t - \beta z - \theta_\eta). \quad (8-91)$$

Instantaneous(瞬态) Poynting Vector



$$\begin{aligned}\mathcal{P}(z, t) &= \mathbf{E}(z, t) \times \mathbf{H}(z, t) = \mathcal{R}e[\mathbf{E}(z)e^{j\omega t}] \times \mathcal{R}e[\mathbf{H}(z)e^{j\omega t}] \\ &= \mathbf{a}_z \frac{E_0^2}{|\eta|} e^{-2\alpha z} \cos(\omega t - \beta z) \cos(\omega t - \beta z - \theta_\eta) \\ &= \mathbf{a}_z \frac{E_0^2}{2|\eta|} e^{-2\alpha z} [\cos \theta_\eta + \cos(2\omega t - 2\beta z - \theta_\eta)].\end{aligned}$$

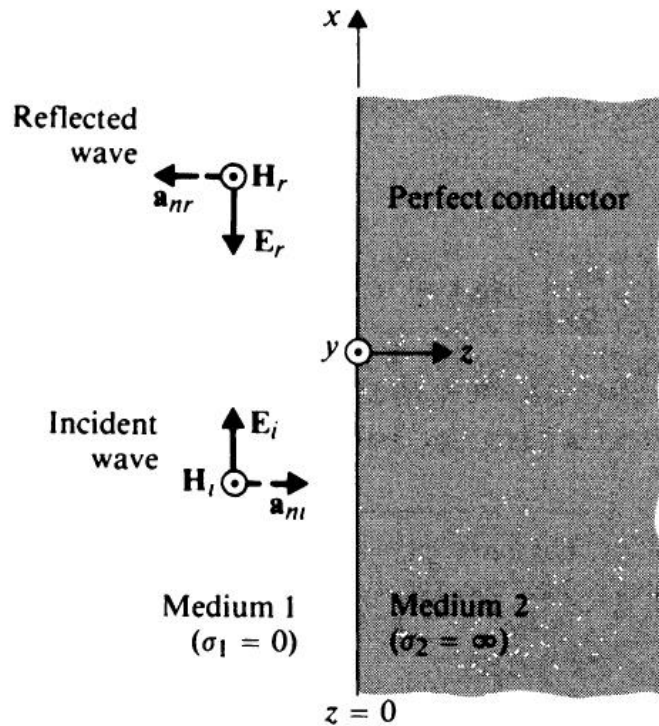
Time-average (时间平均) Poynting Vector

$$\mathcal{P}_{av}(z) = \frac{1}{T} \int_0^T \mathcal{P}(z, t) dt = \mathbf{a}_z \frac{E_0^2}{2|\eta|} e^{-2\alpha z} \cos \theta_\eta \quad (\text{W/m}^2),$$

General Form $\mathcal{P}_{av} = \frac{1}{2} \mathcal{R}e(\mathbf{E} \times \mathbf{H}^*)$

§ 8-6 Normal Incidence at a Plane Conducting Boundary (A uniform plane wave is incident normally on a perfect conductor)

Lossless medium
(conductivity $\sigma=0$)



Perfect conductor
(conductivity $\sigma=+\infty$)

$$\mathbf{E}_i(z) = \mathbf{a}_x E_{i0} e^{-j\beta_1 z},$$

$$\mathbf{H}_i(z) = \mathbf{a}_y \frac{E_{i0}}{\eta_1} e^{-j\beta_1 z},$$

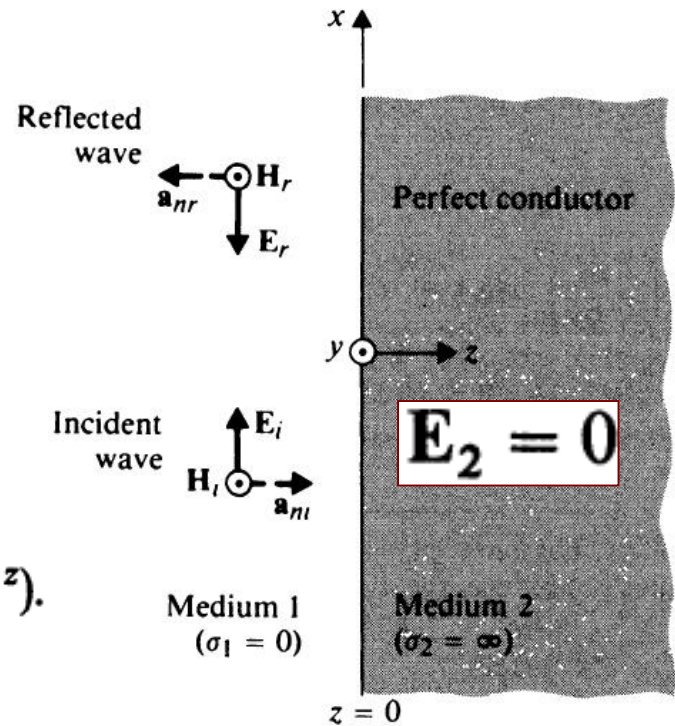
Total electric field in medium 1:

Incident field $\mathbf{E}_i$: $\mathbf{E}_i(z) = \mathbf{a}_x E_{i0} e^{-j\beta_1 z}$,

Reflected field $\mathbf{E}_r$: $\mathbf{E}_r(z) = \mathbf{a}_x E_{r0} e^{+j\beta_1 z}$,

Total field $\mathbf{E}_1$:

$$\mathbf{E}_1(z) = \mathbf{E}_i(z) + \mathbf{E}_r(z) = \mathbf{a}_x (E_{i0} e^{-j\beta_1 z} + E_{r0} e^{+j\beta_1 z}).$$



boundary continuity condition, $E_1(0) = E_2(0)$

$$\mathbf{E}_1(0) = \mathbf{a}_x (E_{i0} + E_{r0}) = \mathbf{E}_2(0) = 0, \quad \Rightarrow \quad E_{r0} = -E_{i0}$$

Total field $\mathbf{E}_1$: $\mathbf{E}_1(z) = \mathbf{a}_x E_{i0} (e^{-j\beta_1 z} - e^{+j\beta_1 z})$

(standing wave; 驻波) $= -\mathbf{a}_x j 2 E_{i0} \sin \beta_1 z$.

Total magnetic field in medium 1:

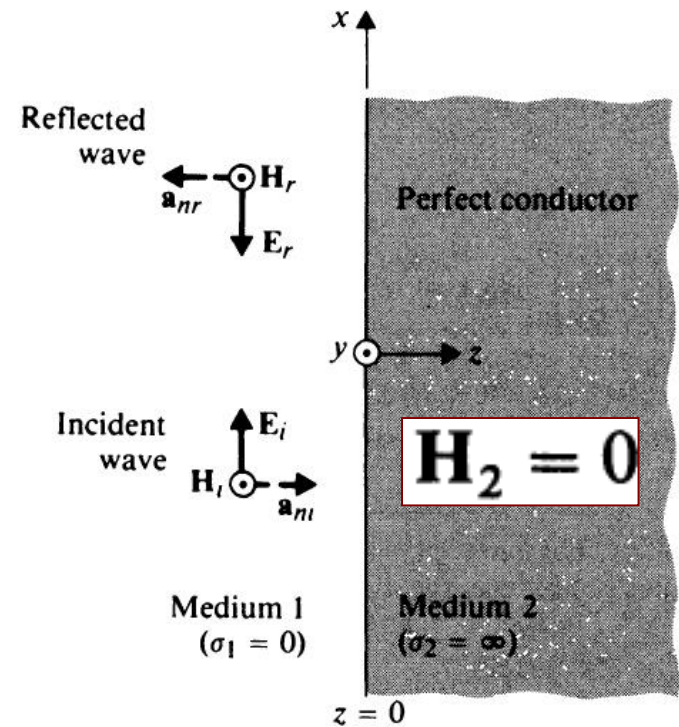
Incident field $\mathbf{H}_i$: $\mathbf{H}_i(z) = \mathbf{a}_y \frac{E_{i0}}{\eta_1} e^{-j\beta_1 z}$

Reflected field $\mathbf{H}_r$:

$$\mathbf{H}_r(z) = \mathbf{a}_y \frac{E_{i0}}{\eta_1} e^{+j\beta_1 z}$$

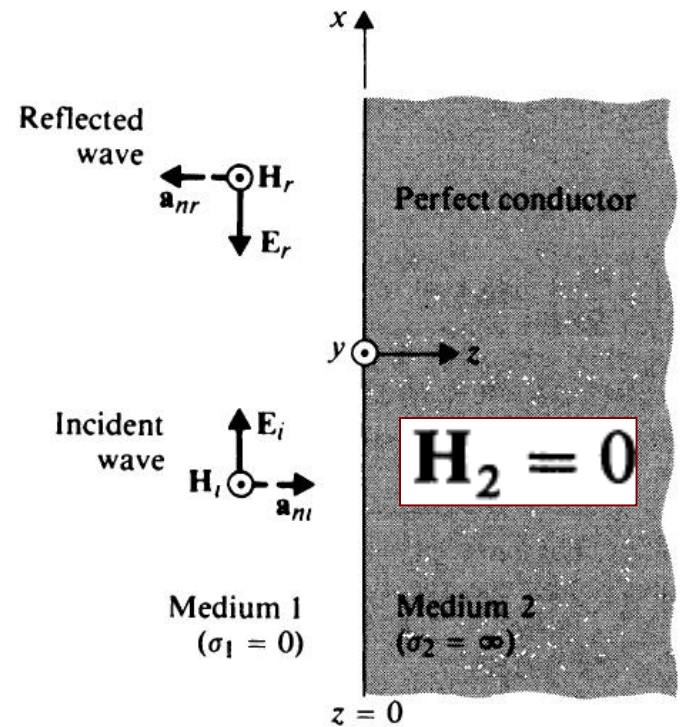
Total field $\mathbf{H}_1$:

$$\mathbf{H}_1(z) = \mathbf{H}_i(z) + \mathbf{H}_r(z) = \mathbf{a}_y 2 \frac{E_{i0}}{\eta_1} \cos \beta_1 z.$$



$$\begin{aligned}\mathbf{E}_1(z) &= \mathbf{a}_x E_{i0} (e^{-j\beta_1 z} - e^{+j\beta_1 z}) \\ &= -\mathbf{a}_x \underline{j} 2 E_{i0} \sin \beta_1 z.\end{aligned}$$

$$\mathbf{H}_1(z) = \mathbf{H}_i(z) + \mathbf{H}_r(z) = \mathbf{a}_y 2 \frac{E_{i0}}{\eta_1} \cos \beta_1 z.$$



$$\mathcal{P}_{av} = \frac{1}{2} \Re(\mathbf{E} \times \mathbf{H}^*) = \frac{1}{2} \times \Re(\text{纯虚数}) = 0$$

$$\epsilon_0, \mu_0 \quad (\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}})$$

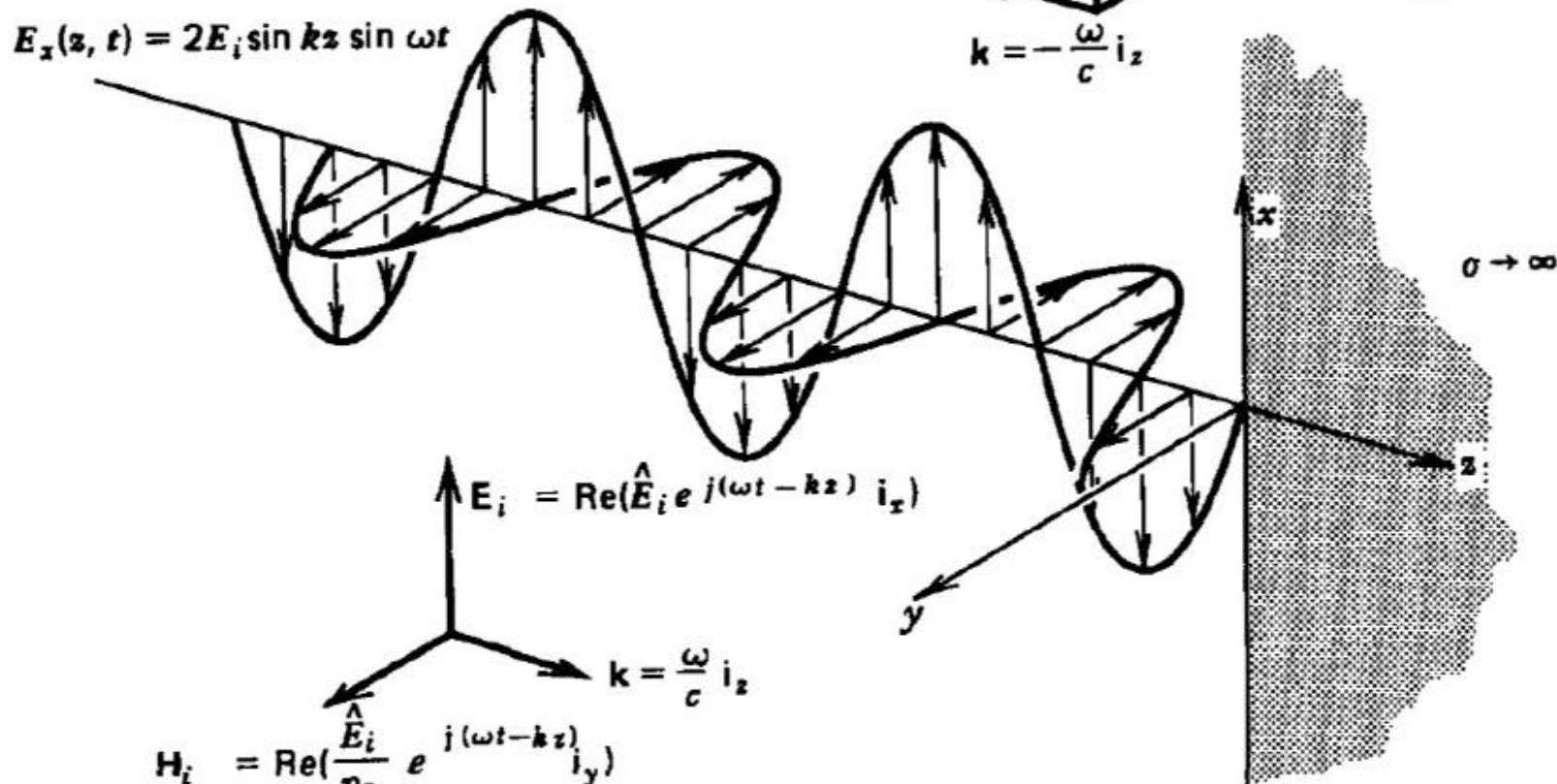
$$H_y(z, t) = \frac{2E_i}{\eta_0} \cos kz \cos \omega t$$

$$E_x(z, t) = 2E_i \sin kz \sin \omega t$$

$$\begin{aligned} E_r &= \text{Re}(\hat{E}_r e^{j(\omega t + kz)} i_x) \\ H_r &= \text{Re}\left(-\frac{\hat{E}_r}{\eta_0} e^{j(\omega t + kz)} i_y\right) \\ k &= -\frac{\omega}{c} i_z \end{aligned}$$

$$\begin{aligned} E_i &= \text{Re}(\hat{E}_i e^{j(\omega t - kz)} i_x) \\ H_i &= \text{Re}\left(\frac{\hat{E}_i}{\eta_0} e^{j(\omega t - kz)} i_y\right) \\ k &= \frac{\omega}{c} i_z \end{aligned}$$

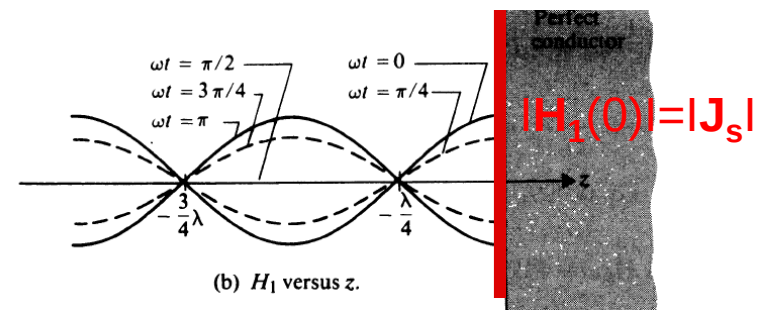
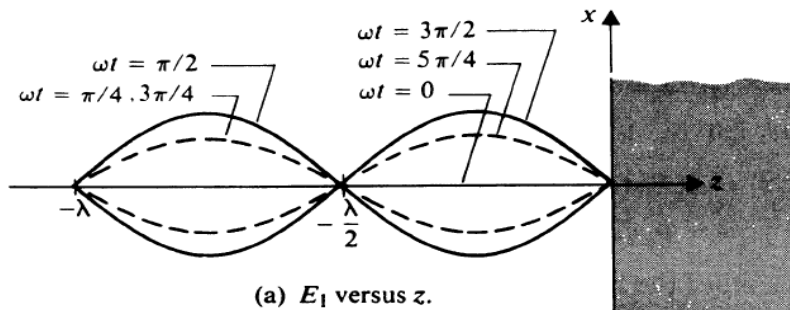
$$\sigma \rightarrow \infty$$



Standing waves (驻波)

$$\mathbf{E}_1(z, t) = \Re[\mathbf{E}_1(z)e^{j\omega t}] = \mathbf{a}_x 2E_{i0} \sin \beta_1 z \sin \omega t,$$

Zeros of $\mathbf{E}_1(z, t)$
Maxima of $\mathbf{H}_1(z, t)$ } occur at $\beta_1 z = -n\pi$, or $z = -n\frac{\lambda}{2}$, $n = 0, 1, 2, \dots$



Standing waves: superimposition of two waves traveling in the opposite directions.

$$\mathbf{H}_1(z, t) = \Re[\mathbf{H}_1(z)e^{j\omega t}] = \mathbf{a}_y 2 \frac{E_{i0}}{\eta_1} \cos \beta_1 z \cos \omega t.$$

Maxima of $\mathbf{E}_1(z, t)$
Zeros of $\mathbf{H}_1(z, t)$ } occur at $\beta_1 z = -(2n + 1)\frac{\pi}{2}$, or $z = -(2n + 1)\frac{\lambda}{4}$, $n = 0, 1, 2, \dots$

